

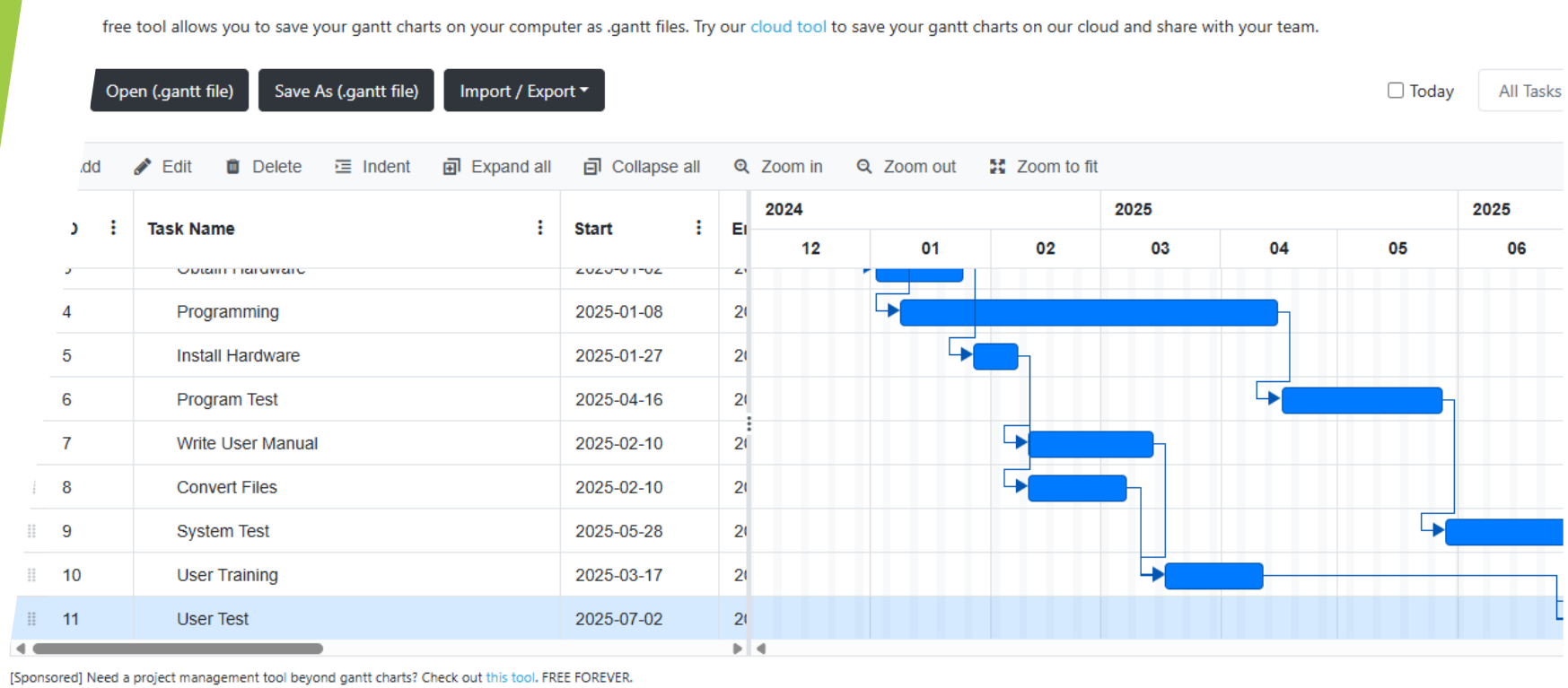
Mackinac Gnomes Order Management System

Semester Project & Systems Analysis

Presented By: Neal Kauffman

Project Overview

- The Challenge: Legacy manual spreadsheet is inefficient for a growing company
- The Goal: Transition to Cloud-Native, Automated order entry and tracking application
- Key Objectives:
  - Centralize Data
  - Automate Shipping Workflows
  - Improve Reporting Accuracy



Project Management

To Review:

Name	Start	Finish	Duration	%Comp
Create Project Shell Documentation	2/16/2026	2/16/2026	1 day	
Survey Users	2/17/2026	2/23/2026	5 days	
Create a preliminary plan	2/17/2026	2/18/2026	2 days	
Produce Data Flow Diagrams	2/24/2026	2/27/2026	4 days	
Produce UML Diagrams	2/19/2026	2/24/2026	4 days	
Receive Feedback on Diagrams	3/2/2026	3/4/2026	3 days	
Create UI Mockups	3/5/2026	3/12/2026	6 days	
Produce ERD Diagrams	3/5/2026	3/11/2026	5 days	
Produce a System Architecture	3/12/2026	3/17/2026	4 days	
Develop Presentation	3/5/2026	3/24/2026	14 days	
Record Presentation	3/25/2026	3/25/2026	1 day	
Present to Management	3/25/2026	3/25/2026	1 day	
Review Requirements with Stakeholders	2/24/2026	2/25/2026	2 days	
Revise Diagrams after Feedback	3/5/2026	3/9/2026	3 days	
Warranty Support	3/26/2026	4/6/2026	8 days	

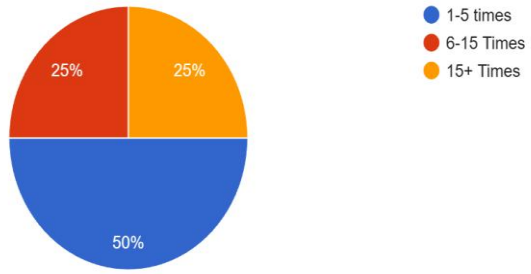
- Key Phases:
  - Analysis and Workflow Strategy
  - Data Architecture & UI/UX Design
  - Infrastructure & Handoff

Requirements Gathering

Survey Results

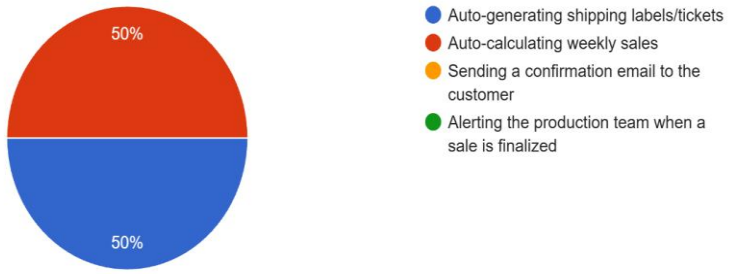
On an average day, how many times do you access the spreadsheet?

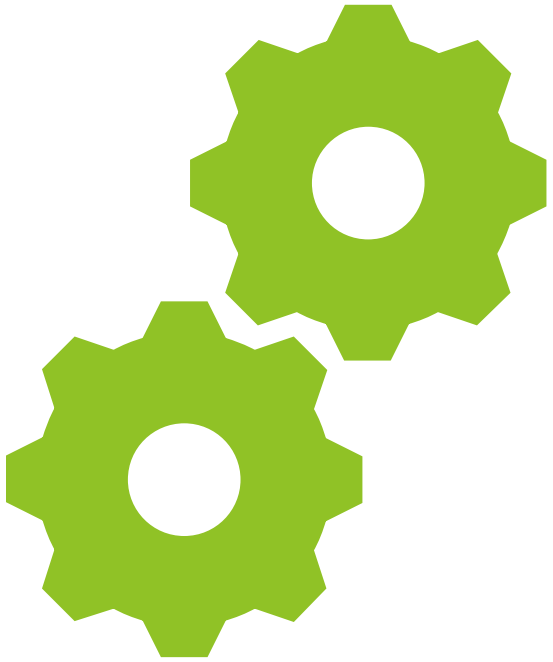
4 responses



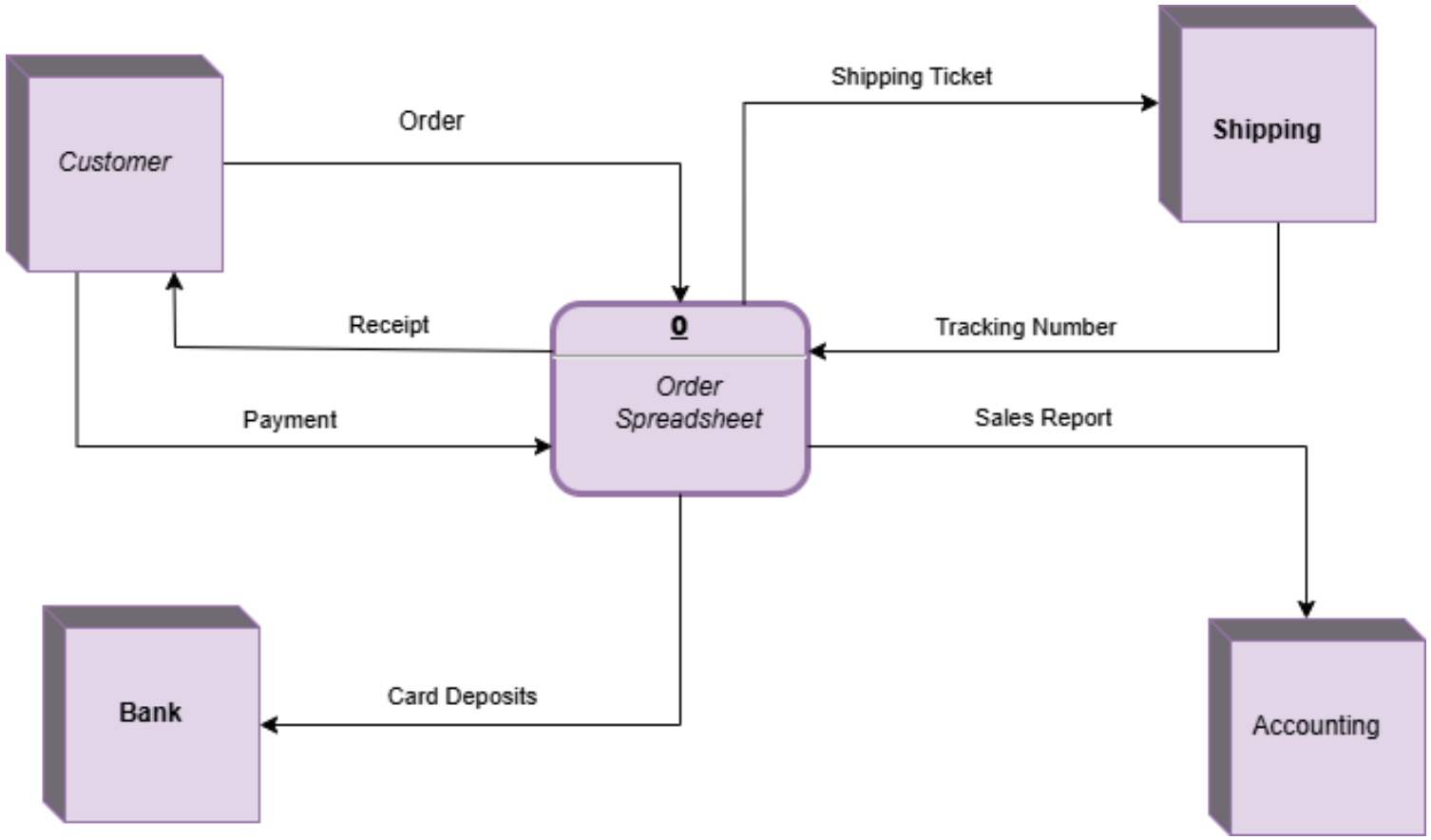
If the new system could automate ONE task, which would be the most helpful?

4 responses



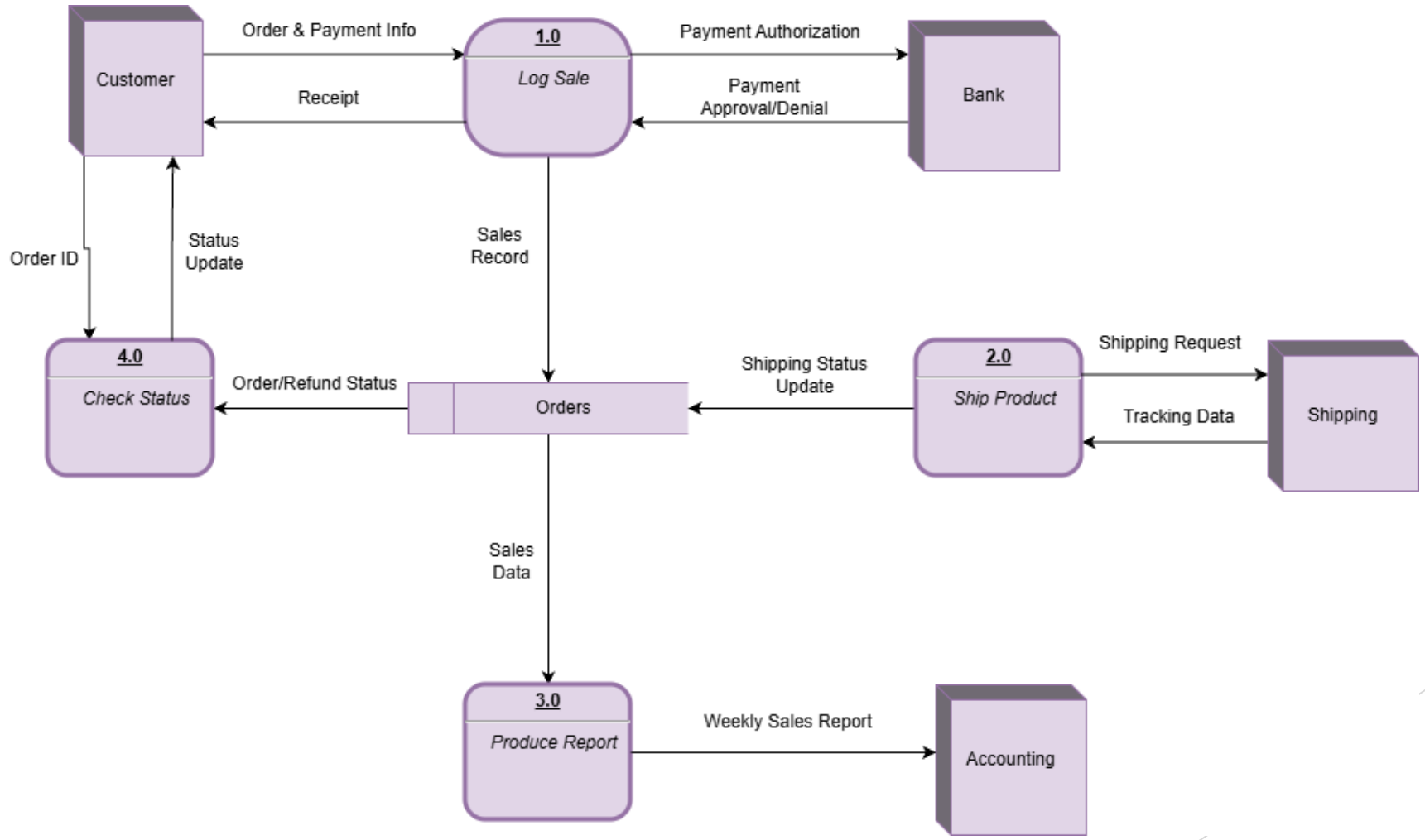


Process Modeling



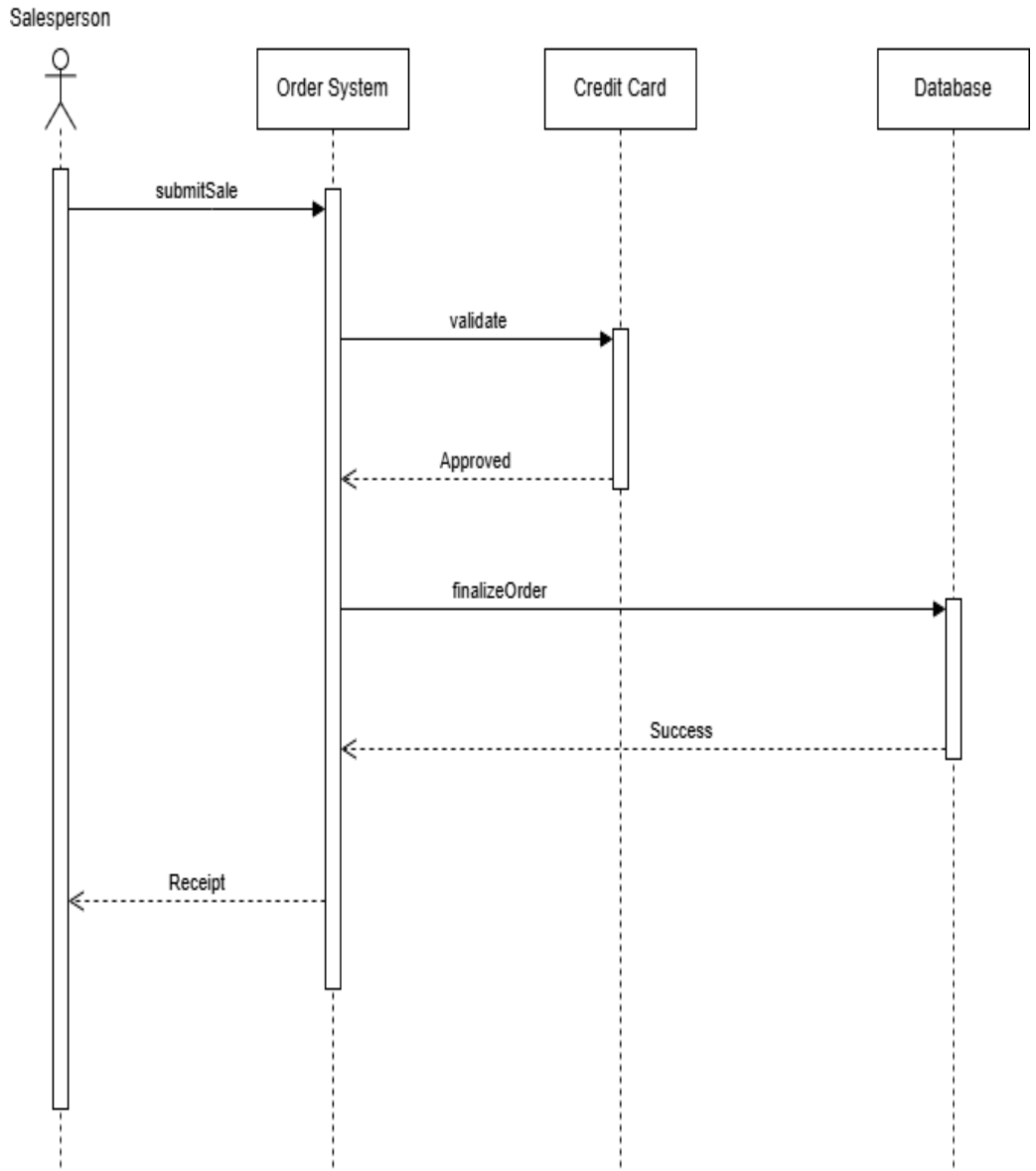
Old System Data Flow

Proposed New System Data Flow

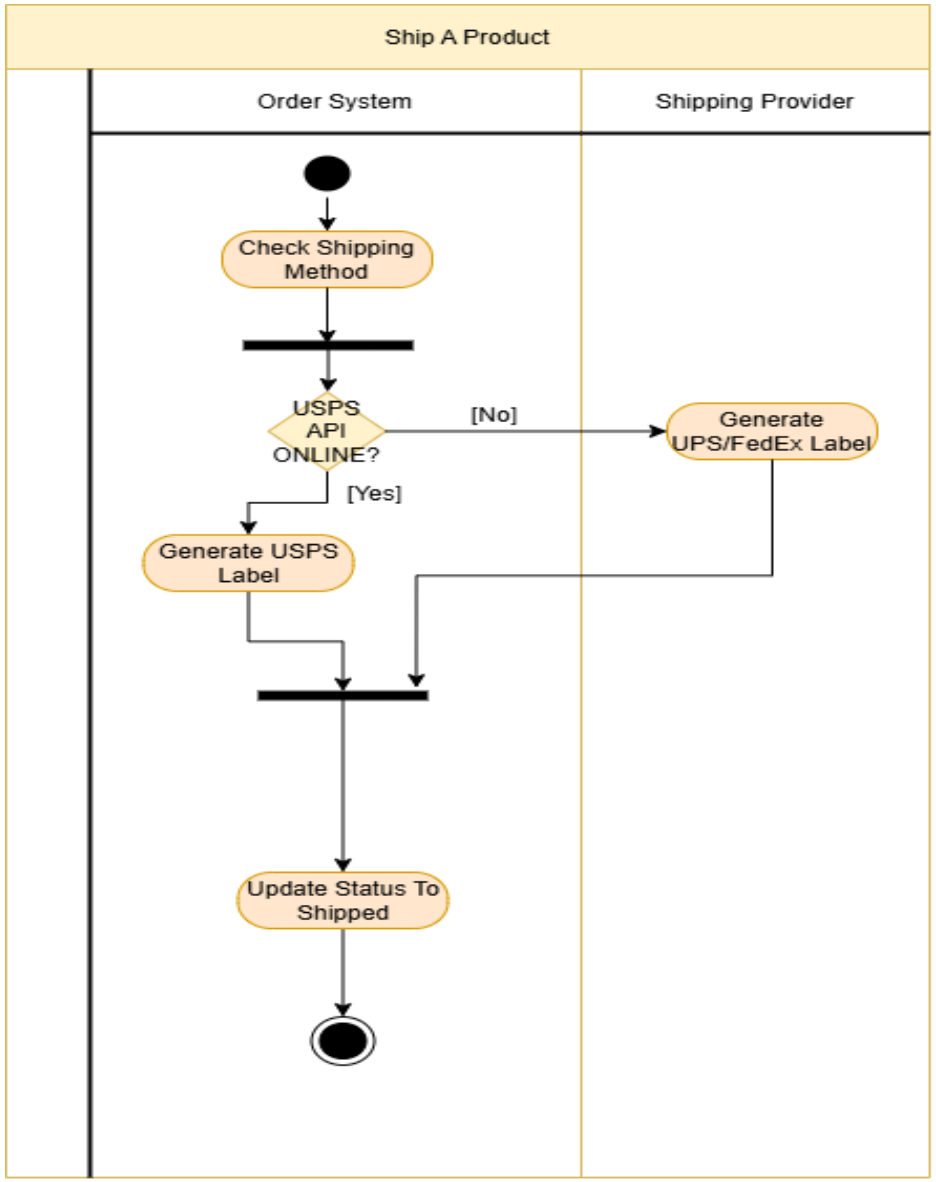


Label	LOG SALE
Entry Type	Process
Description	Captures and validates transactional data for a single gnome sale
Process #	1.0
Description	Input data flows: ORDER & PAYMENT INFO, PAYMENT APPROVAL/DENIAL Output data flows: PAYMENT AUTHORIZATION, SALES RECORD, RECEIPT For each ORDER & PAYMENT INFO: Output PAYMENT AUTHORIZATION to Bank If PAYMENT APPROVAL/DENIAL = Approved THEN Output SALES RECORD to Orders Data Store Output RECEIPT to Customer Else Notify Customer of Payment Error
Notes:	All sales restricted to one gnome per business rule.

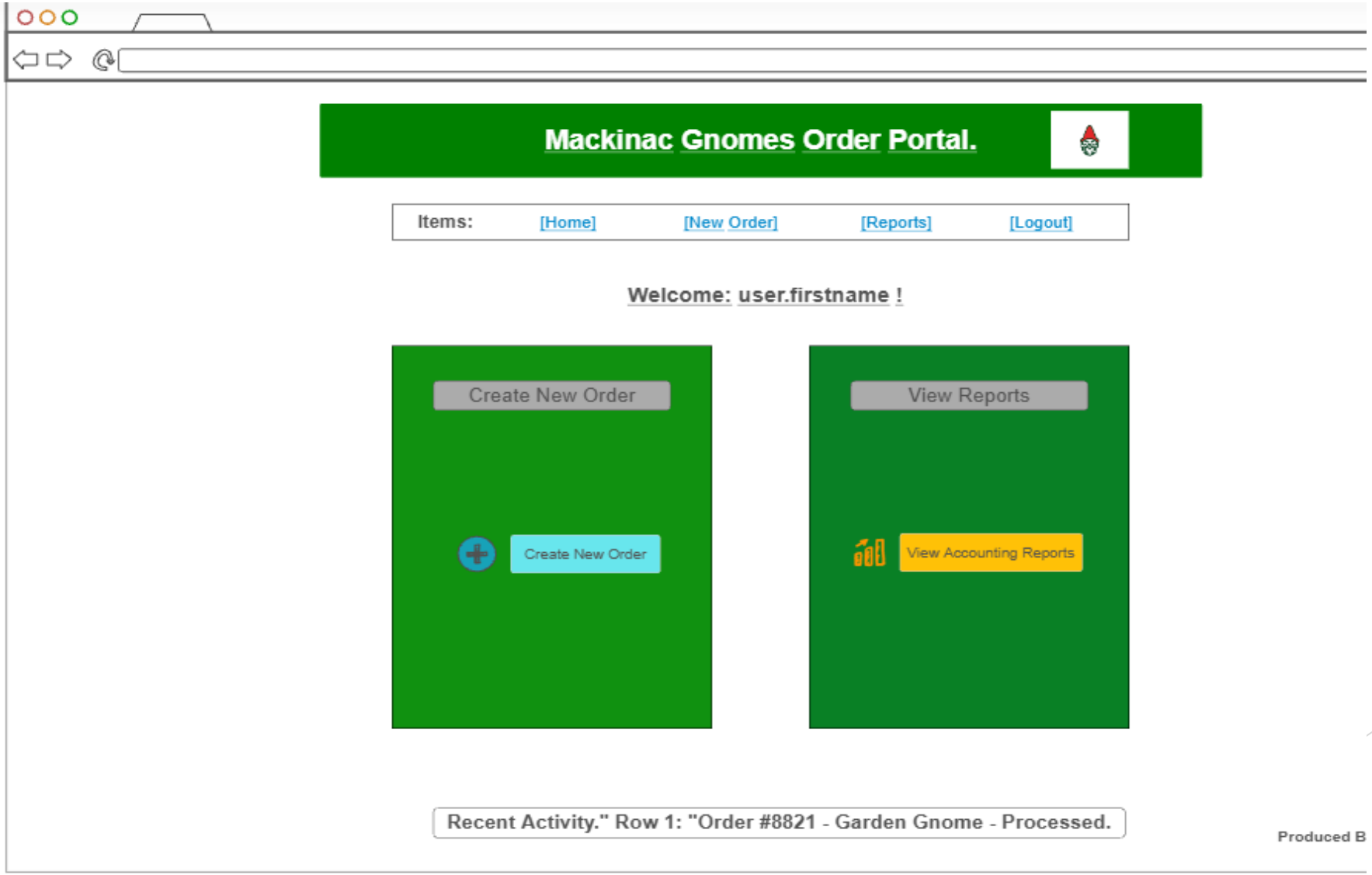
How does it Work?


Salesperson
Order System
Credit Card
Database
submitSale
validate
Approved
finalizeOrder
Success
Receipt

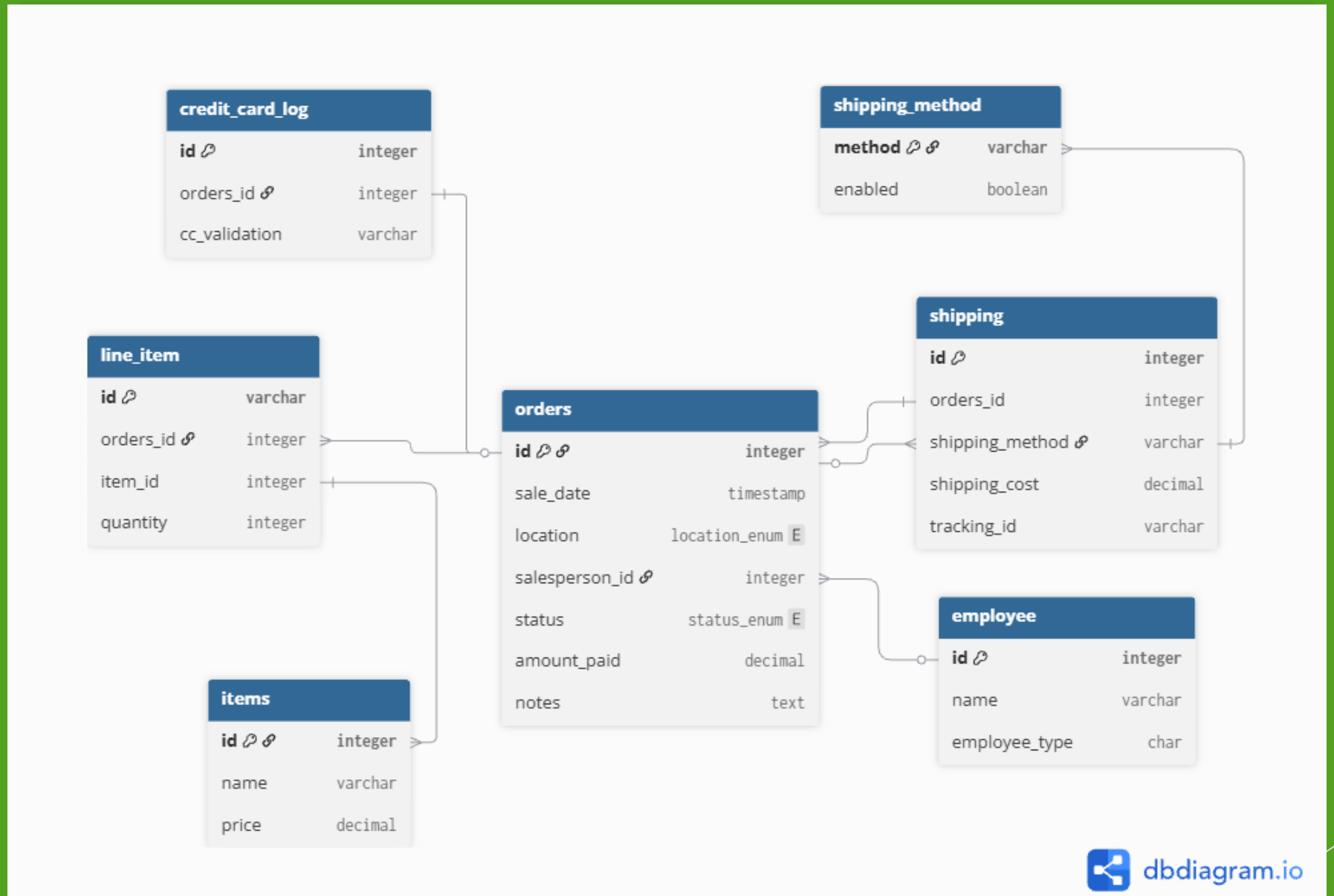
In Sequence


Ship A Product
Order System
Shipping Provider
Check Shipping Method
USPS API ONLINE?
[No]
Generate UPS/FedEx Label
[Yes]
Generate USPS Label
Update Status To Shipped

What Will It Look Like?



Behind the Data



```
CREATE TYPE "location_enum" AS ENUM (
  'Tradeshow',
  'Online_eBay',
  'Online_Etsy',
  'Online_Other'
);
CREATE TYPE "status_enum" AS ENUM (
  'Delivered',
  'Paid',
  'Returned',
  'Shipped'
);
CREATE TABLE "items" (
  "id" integer PRIMARY KEY,
  "name" varchar,
  "price" decimal
);
CREATE TABLE "shipping_method" (
  "method" varchar PRIMARY KEY,
  "enabled" boolean
);
CREATE TABLE "employee" (
  "id" integer PRIMARY KEY,
  "name" varchar,
  "employee_type" char
);
CREATE TABLE "orders" (
  "id" integer PRIMARY KEY,
  "sale_date" timestamp,
  "location" location_enum,
  "salesperson_id" integer,
  "status" status_enum,
  "amount_paid" decimal,
  "notes" text
);
CREATE TABLE "credit_card_log" (
  "id" integer PRIMARY KEY,
  "orders_id" integer,
  "cc_validation" varchar
);
CREATE TABLE "shipping" (
  "id" integer PRIMARY KEY,
  "orders_id" integer,
  "shipping_method" varchar,
  "shipping_cost" decimal,
  "tracking_id" varchar
);
CREATE TABLE "line_item" (
  "id" varchar PRIMARY KEY,
  "orders_id" integer,
  "item_id" integer,
  "quantity" integer
);
ALTER TABLE "credit_card_log" ADD FOREIGN KEY ("orders_id") REFERENCES "orders" ("id") DEFERRABLE INITIALLY IMMEDIATE;

ALTER TABLE "orders" ADD FOREIGN KEY ("salesperson_id") REFERENCES "employee" ("id") DEFERRABLE INITIALLY IMMEDIATE;

ALTER TABLE "line_item" ADD FOREIGN KEY ("orders_id") REFERENCES "orders" ("id") DEFERRABLE INITIALLY IMMEDIATE;

ALTER TABLE "items" ADD FOREIGN KEY ("id") REFERENCES "line_item" ("item_id") DEFERRABLE INITIALLY IMMEDIATE;

ALTER TABLE "shipping" ADD FOREIGN KEY ("shipping_method") REFERENCES "orders" ("id") DEFERRABLE INITIALLY IMMEDIATE;

ALTER TABLE "orders" ADD FOREIGN KEY ("id") REFERENCES "shipping" ("orders_id") DEFERRABLE INITIALLY IMMEDIATE;

ALTER TABLE "shipping_method" ADD FOREIGN KEY ("method") REFERENCES "shipping" ("shipping_method") DEFERRABLE INITIALLY IMMEDIATE;

```

3 Write SQL Query Ctrl-Alt-3

```sql
-- Generate similar to spreadsheet
SELECT 'Neal Kauffman' Generated_By, o.id, o.sale_date, o.location,
e.name, i.name, o.status,o.amount_paid,
ccl.cc_validation, s.shipping_method, s.shipping_cost, o.notes
FROM orders o
left join line_item li on li.order_id = o.id
left join items i on i.id = li.item_id
left join credit_card_log ccl on ccl.order_id = o.id
left join shipping s on s.order_id = o.id
left join employee e on e.id = o.salesperson_id;

```

1 error

Run Selected Ctrl-↵ ✓ 6 rows returned. Executed in 3.620989 ms.

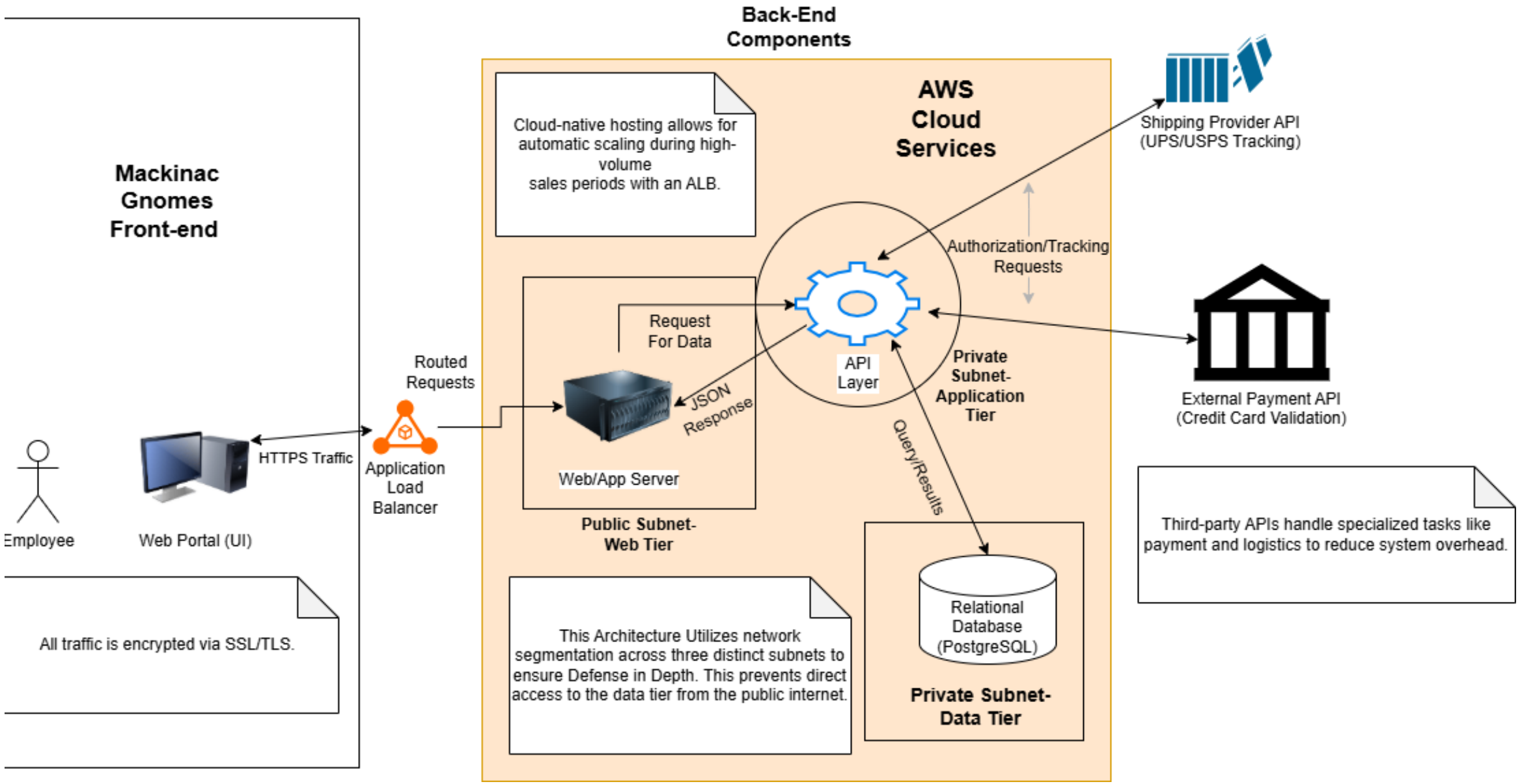
Query Result Download CSV

generated_by	id	sale_date	location	name	name
Neal Kauffman	10001	2025-01-01 00:00:00	Tradeshow	Jim	Garden Gnom
Neal Kauffman	10002	2025-01-01 00:00:00	Online_eBay	Jim	Plastic Gnome
Neal Kauffman	10002	2025-01-01 00:00:00	Online_eBay	Jim	Plastic Gnome
Neal Kauffman	10003	2025-01-01 00:00:00	Online_eBay	*(null)*	Plastic Gnome
Neal Kauffman	10004	2025-01-02 00:00:00	Tradeshow	Sheila	Garden Gnom
Neal Kauffman	10005	2025-01-03 00:00:00	Tradeshow	Mark	Garden Gnom

SQL In Action



System Architecture


Back-End Components
Mackinac Gnomes Front-end
AWS Cloud Services
Cloud-native hosting allows for automatic scaling during high-volume sales periods with an ALB.
Shipping Provider API (UPS/USPS Tracking)
Authorization/Tracking Requests
Request For Data
Routed Requests
API Layer
Private Subnet-Application Tier
External Payment API (Credit Card Validation)
JSON Response
Query/Results
Employee
Web Portal (UI)
HTTPS Traffic
Application Load Balancer
Web/App Server
Public Subnet-Web Tier
Third-party APIs handle specialized tasks like payment and logistics to reduce system overhead.
All traffic is encrypted via SSL/TLS.
This Architecture Utilizes network segmentation across three distinct subnets to ensure Defense in Depth. This prevents direct access to the data tier from the public internet.
Relational Database (PostgreSQL)
Private Subnet-Data Tier

Expected Outcomes & Conclusion



Scalability



Defense In Depth



Automation = ROI ↑